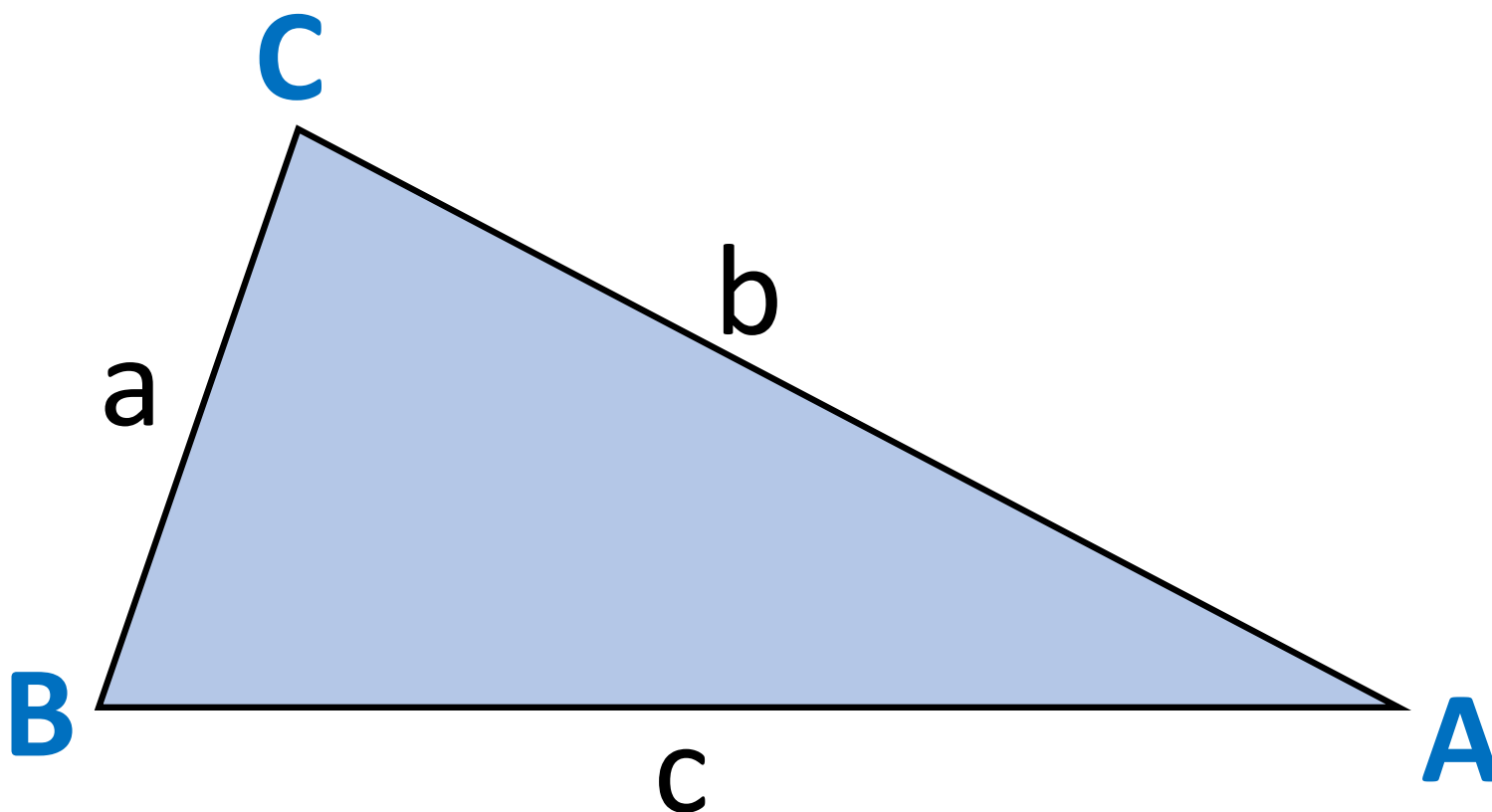


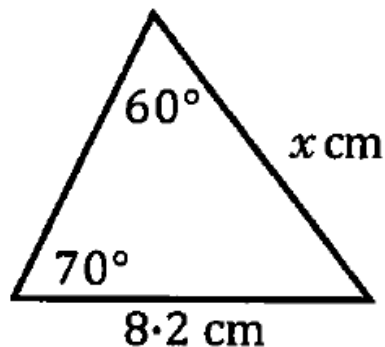
# Sine and Cosine Rules



Sine rule:

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

# Examples

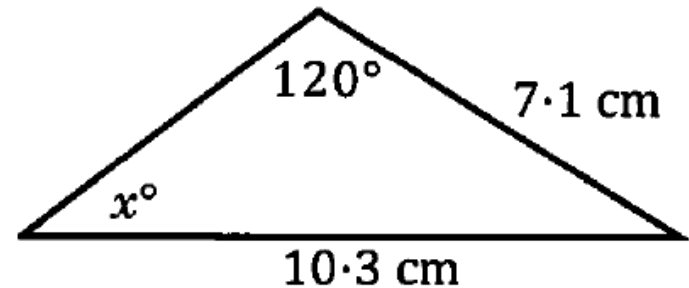


By the sine rule

$$\frac{x}{\sin 70^\circ} = \frac{8.2}{\sin 60^\circ}$$

Multiply by  $\sin 70^\circ$  to isolate  $x$

$$\begin{aligned} x &= \frac{8.2 \sin 70^\circ}{\sin 60^\circ} \\ &= 8.9 \text{ (to 1 dp)} \end{aligned}$$



By the sine rule

$$\frac{10.3}{\sin 120^\circ} = \frac{7.1}{\sin x^\circ}$$

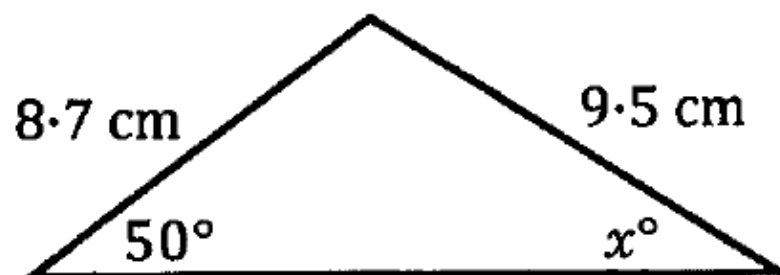
Multiply by  $(\sin x^\circ)(\sin 120^\circ)$

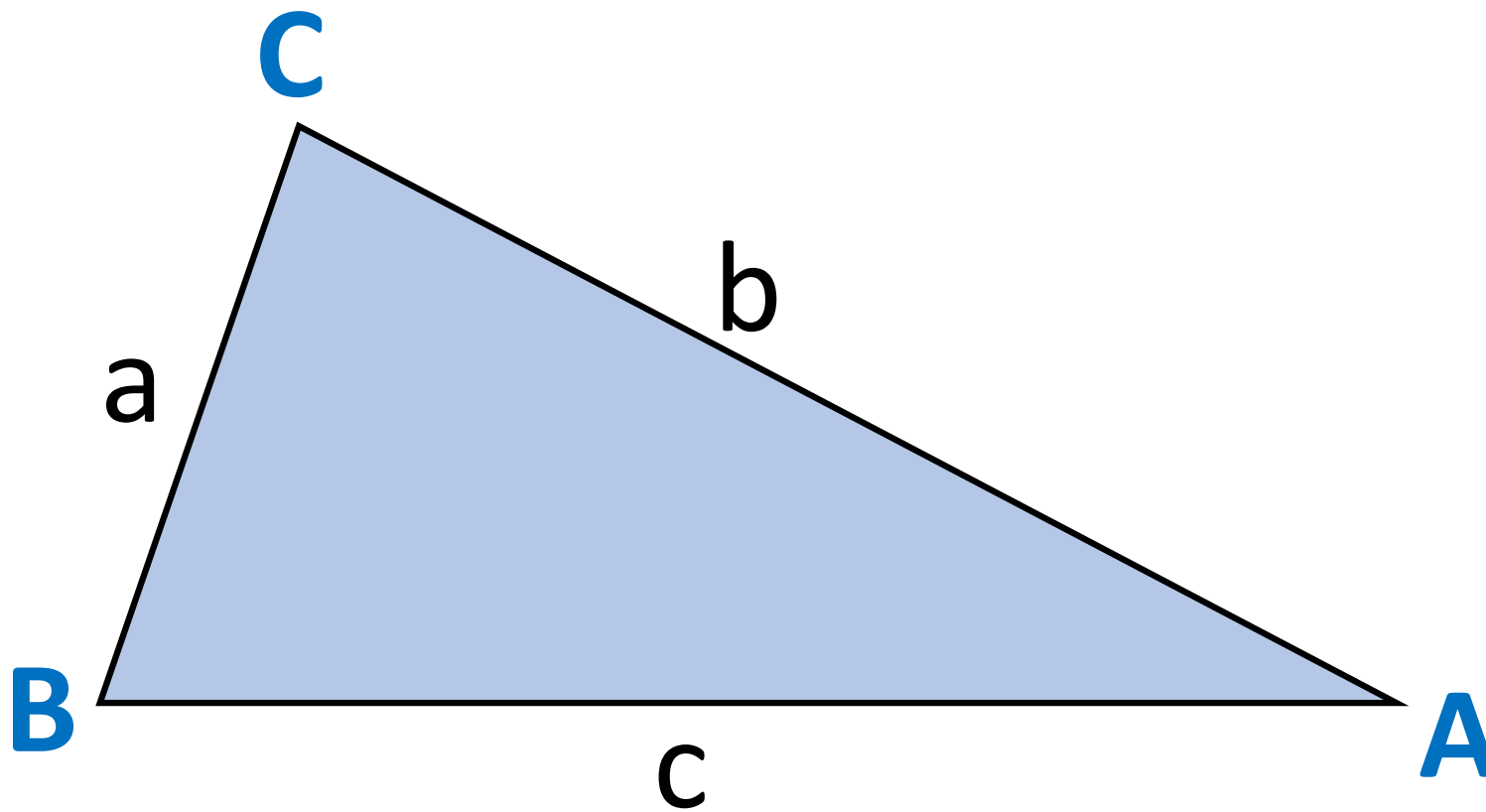
$$10.3 \sin x^\circ = 7.1 \sin 120^\circ$$

$$\sin x^\circ \approx 0.5970$$

$$x = 36.7 \text{ (to 1 dp)}$$

Find the value of  $x$  in the triangle shown on the right.  
(Give the answer correct to one decimal place.)





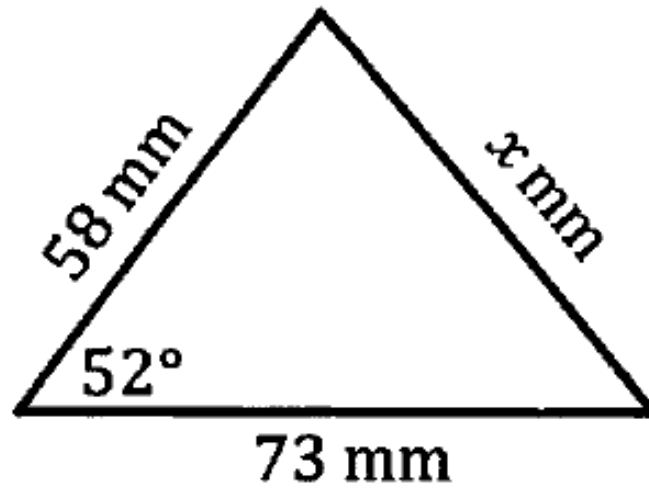
Cosine rule:

$$c^2 = a^2 + b^2 - 2 ab \cos C$$

$$b^2 = a^2 + c^2 - 2 ac \cos B$$

$$a^2 = b^2 + c^2 - 2 bc \cos A$$

# Examples



The sketch on the right shows a system of three triangles with lengths and angles as indicated.  
BAE is a straight line.  
Find the length of CD.

